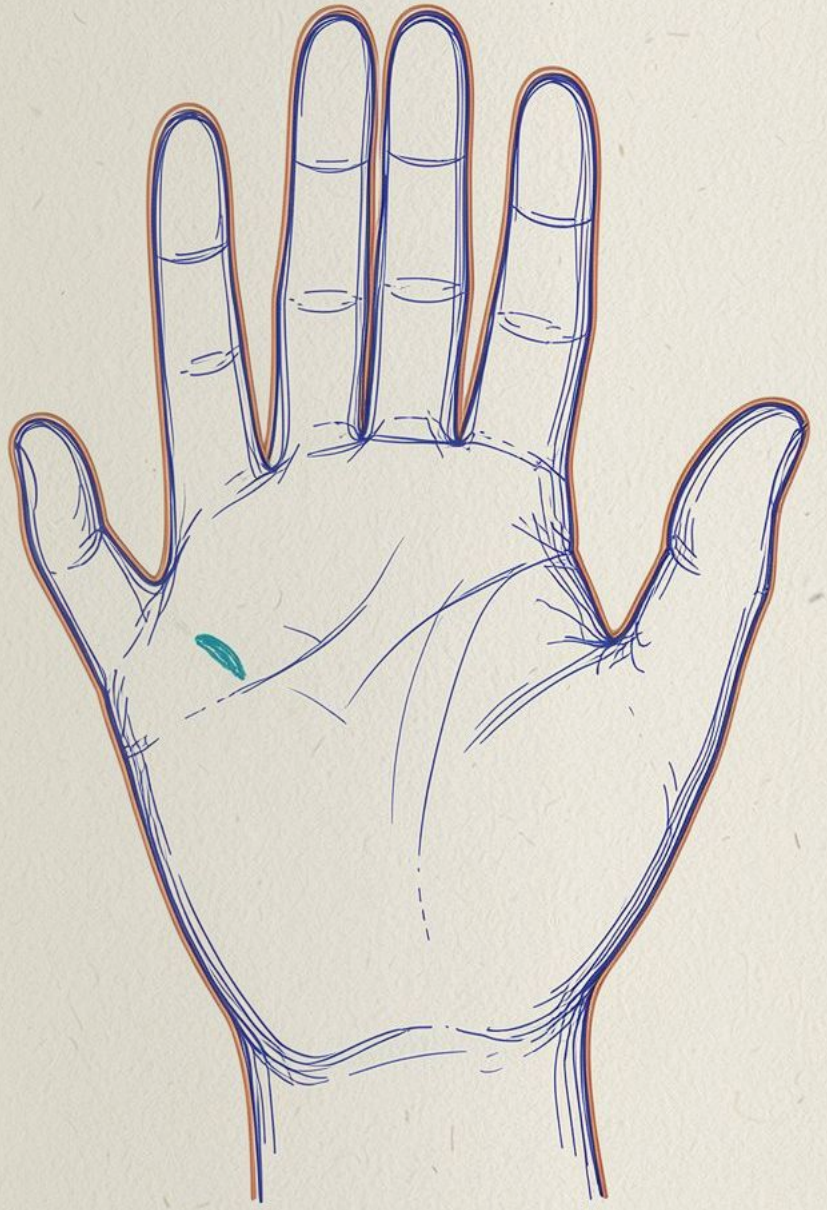
DEEP LEARNING · IMAGE CLASSIFICATION

Hand Gesture Recognition Using CNN

An end-to-end workflow from labeled hand images to reusable prediction artifacts.

Google Colab · Shafaat Ullah



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PROJECT OVERVIEW

From a hand image to a predicted gesture

Build an end-to-end image-classification workflow using a Convolutional Neural Network.

Project type: deep learning and image classification · Platform: Google Colab · Author: Shafaat Ullah.

Dataset acquisition	**Preprocessing**	**Training**
Evaluation	**Prediction**	**Reusable model outputs**

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SCOPE

Project objectives define the full workflow

01

Download and inspect

Download and inspect the LeapGestRecog dataset programmatically.

02

Prepare consistent inputs

Derive labels from folder names and prepare consistent CNN inputs.

03

Evaluate performance

Train a multi-class classifier and evaluate it with test metrics, a classification report, and a confusion matrix.

04

Save for reuse

Save the trained model, class mapping, and prediction results for reuse.

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The dataset supplies labeled hand images

LeapGestRecog

Downloaded through KaggleHub rather than a manual ZIP upload.

Folder-derived labels

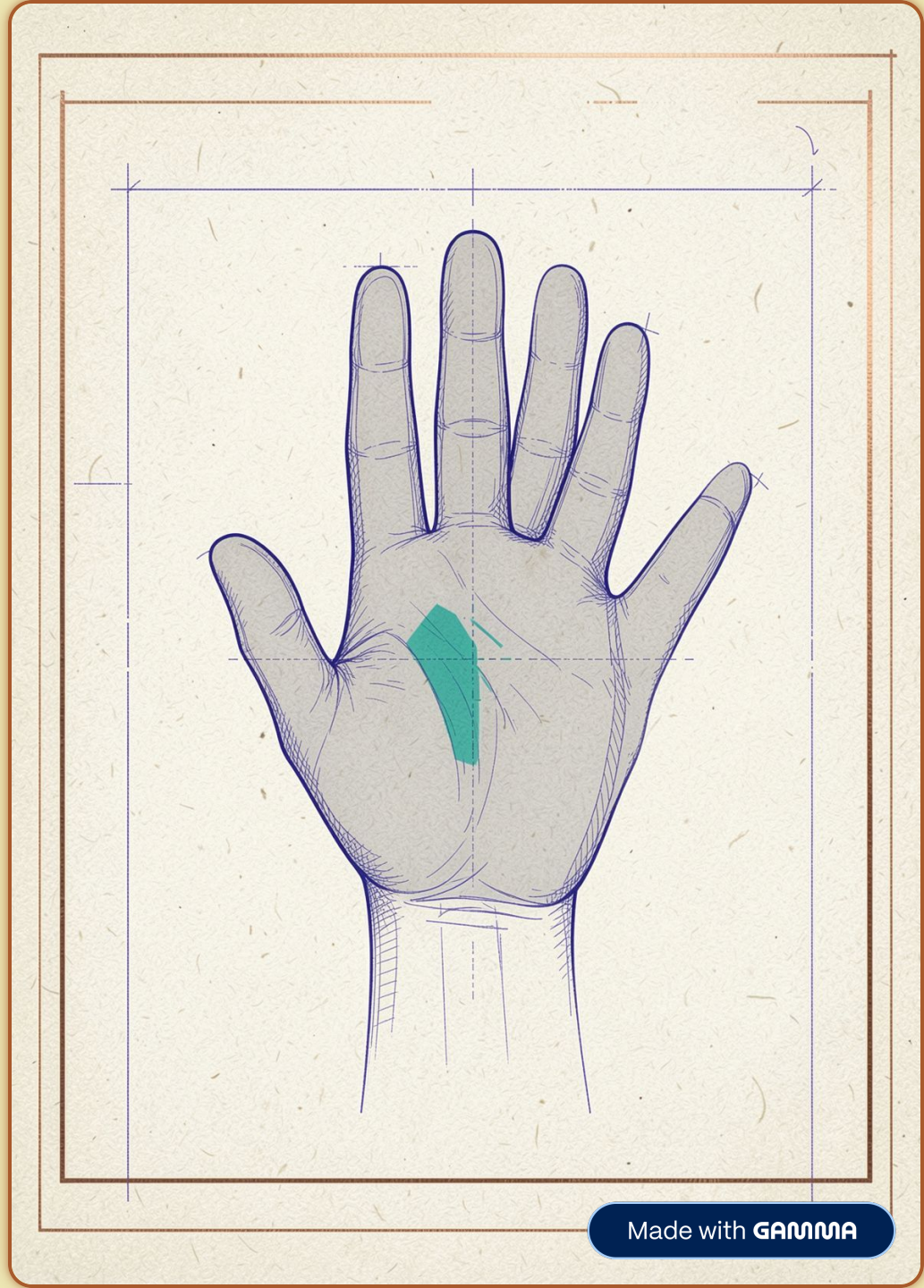
The notebook recursively scans PNG and JPG files and uses each image’s parent folder as its label.

Stratified split

70% training, 15% validation, and 15% testing.

Inspection output

Exact image and class counts are confirmed from the notebook inspection output.



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END-TO-END PIPELINE

The pipeline turns raw images into model-ready batches

1

Download

Download the dataset and discover image paths.

2

Split

Create labels and stratified train, validation, and test subsets.

3

Batch

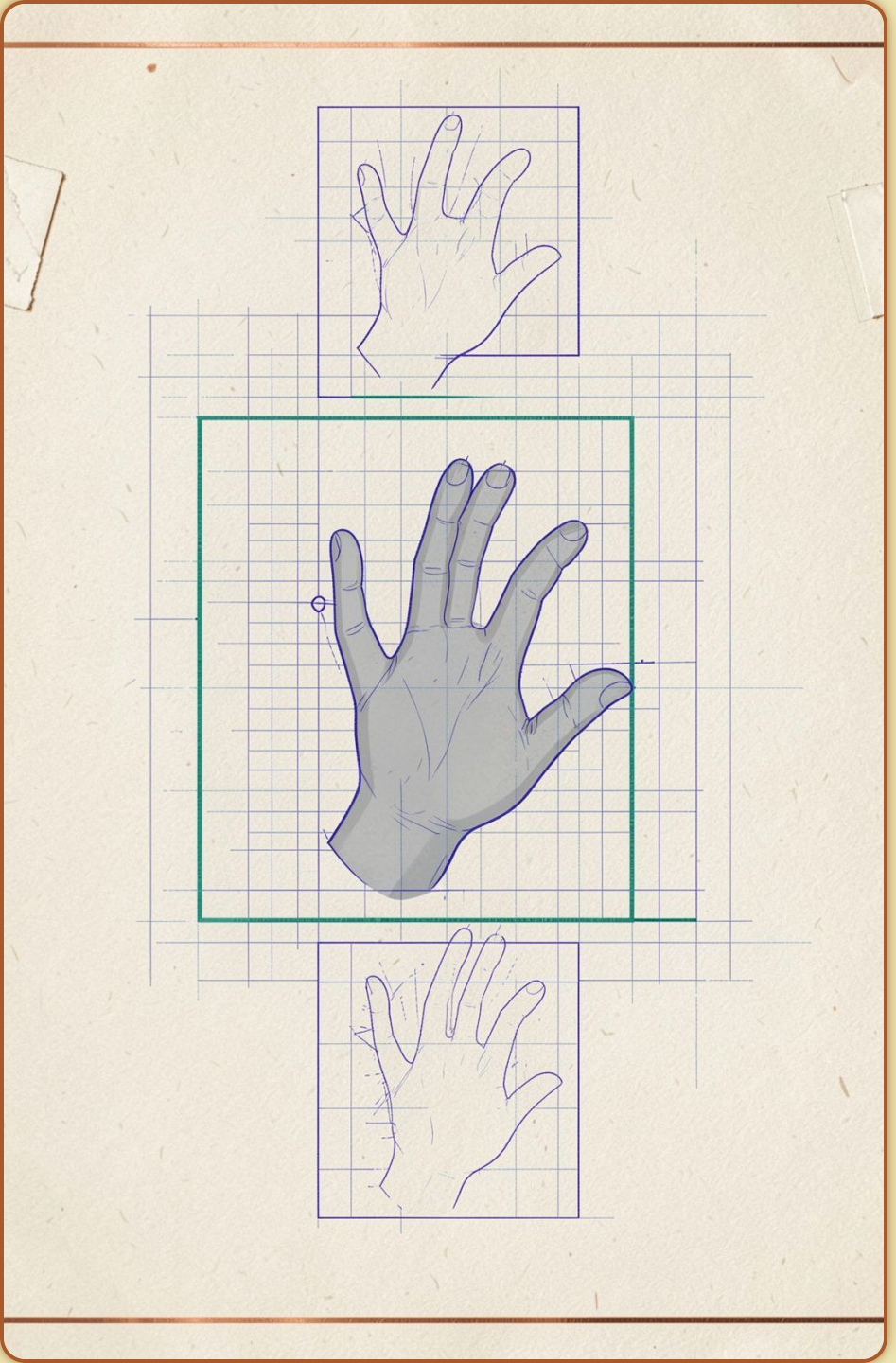
Decode, grayscale, resize, normalize, batch, and prefetch images.

4

Export

Train the CNN, evaluate the held-out test set, then export predictions and artifacts.

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INPUT STANDARDIZATION

Preprocessing standardizes every input

01

Decode

Decode each image into a single grayscale channel.

02

Resize

Resize images to 96 × 96 pixels.

03

Normalize

Convert pixel values to floating point and scale them to the 0–1 range.

04

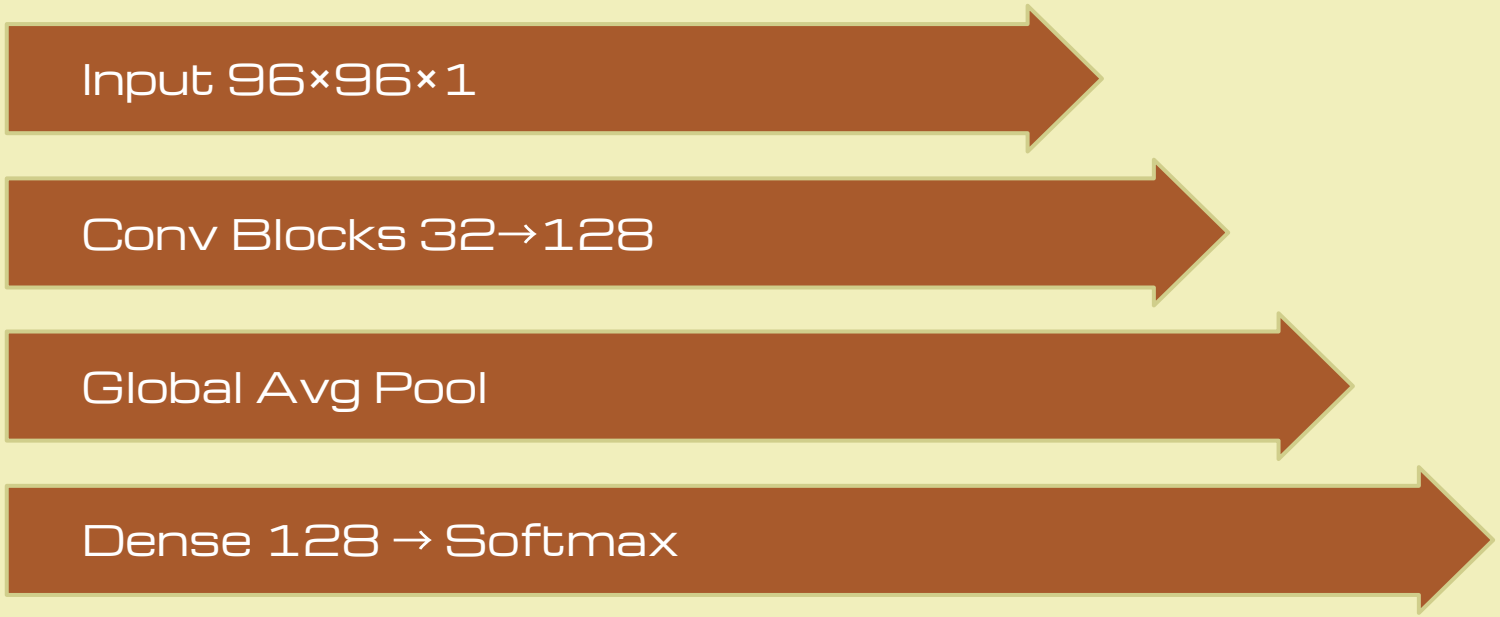
Encode

Map sorted class names to integer IDs for sparse categorical classification
.

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MODEL ARCHITECTURE

The CNN builds visual understanding in stages



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TRAINING CONTROL

Training uses callbacks to protect generalization



Adam optimizer

Initial learning rate: 0.001.



Early stopping

Restores the best weights when validation loss stops improving.



Training budget

Maximum 20 epochs with batch size 32.



Adaptive protection

ReduceLROnPlateau lowers the learning rate; ModelCheckpoint saves the best validation-accuracy model.

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HELD-OUT EVIDENCE

Evaluation must come from the held-out test set

Measure	Run output and purpose
Test accuracy	____ % — Performance on unseen test images.
Test loss	____ — Loss from the completed notebook run.
Image counts	____ / ____ / ____ — Training, validation, and testing image counts from dataset inspection.
Error analysis	Confusion matrix; per-class precision, recall, and F1-score.

Do not substitute training accuracy for unseen-data performance.

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REUSABLE OUTPUTS

The project produces reusable prediction artifacts

hand_gesture_recognition.keras

Saved TensorFlow/Keras model.

gesture_classes.json

Ordered class names matching output indices.

hand_gesture_predictions.csv

Image path, actual label, predicted label, confidence, and correctness flag.

Evaluation exports

Training curves, a confusion matrix, and a classification report.

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